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Algebraic
Multigrid Methods
for Systems

LicenseServer

Binary SAMG Distributions
(Starting with Release 2021)

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Purpose of this Document

This document contains a brief introduction on the installation of SAMG's license server SVD, separately for **Windows** and **Linux / MacOS**.

IMPORTANT:

Please note that only the very basic installation is described here, in order to briefly show FLEXIm-novices how to proceed. Should such a basic installation not meet your requirements, please refer to file [LicenseAdministration.pdf](#) on what to do in different use cases.

We use the following symbols for highlighting important things:

			These symbols are used to draw the reader's attention to particularly important aspects/remarks of this document (order given by increasing importance)
			This symbol is used to mark files that generally require user-dependent adaptations.

Should you need additional help, do not hesitate to contact us via our service e-mail,

samg@scai.fraunhofer.de¹.

Your SAMG Team

¹ **Important:** This e-mail addresses the whole SAMG Team. Therefore, it should not be used to transmit large amounts of data. We will arrange individual solutions on that purpose when needed.

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1 General Remarks

1.1 Contents of this Archive

Since our customers are free to install the SAMG license server on any machine in their network - regardless of its OS, the only prerequisite being that the machine can be reached via TCP/IP from the machines intended to run SAMG applications - the SAMG Release 2021 License Server package contains license server archives, including installation advice, for all platforms SAMG is available for: Windows, Linux and MacOS, (all x64 only).

These OS-specific archives each contain the respective license server tools for SAMG and FLEXIm's License Administration Guide, file [LicenseAdministration.pdf](#), explaining in detail all available options to set up and run the license server.

Contents.txt	Lists the contents of the license server archive
GettingStarted_SAMG_Licserver_Release2021.pdf	This file
LicenseAdministration.pdf	The original FLEXIm v11.18.2.0 License Administration Guide
SVD_flexIm11.18.2.0_x64_lsb.tar.gz	Archive for License Servers on Linux x64
SVD_flexIm11.18.2.0_x64_mac10.tar.gz	Archive for License Servers on MacOS x64
SVD_flexIm11.18.2.0_x64_n6.zip	Archive for License Servers on Windows x64

2 Windows

Below we describe steps to get the FLEXlm license server for SAMG started on Windows. For more information on FLEXlm and its operation, we refer to the **FLEXlm's License Administration Guide** that is provided as part of SAMG's License Server archive for Windows.

2.1 License Server Archive's Content

The [SVD_flexlm11.18.2.0](#) license server archive for Windows contains the following files:

lmgrd.exe	FLEXlm daemon, version 11.18.2.0
SVD.exe	SAMG's vendor daemon; requires lmgrd.exe 11.18.2.0 or higher
lmtools.exe	Windows program for configuring FLEXlm services, starting and stopping the license server, testing license features etc.
lmutil.exe	A corresponding collection of console utilities
LicenseAdministration.pdf	The original FLEXlm v11.18.2.0 License Administration Guide

2.2 Installation of the License Server

Locate the machine that is intended to run SAMG's license server (according to the data you specified to us). Depending on whether or not FLEXlm is already installed on that machine, do the following:

- If FLEXlm is *not* yet installed, unpack the [SVD_flexlm11.18.2.0](#) archive for Windows to a directory on that machine.

Caveat:

Do not install the FLEXlm software and the license file on a network (e.g. CIFS) share. Use a local drive instead. Note that if you run FLEXlm as a service, it may not be allowed to access files on a network file system.



- If FLEXlm *is* already installed, start its [lmtools.exe](#) and go to "Help" → "About" to identify the version number. Then do the following:
 - If the version number is *older* than 11.18.2.0, replace [lmgrd.exe](#), [lmtools.exe](#) and [lmutil.exe](#) with their corresponding newer versions provided in the archive. In addition, put [SVD.exe](#) into the same directory in which [lmgrd.exe](#) resides.
 - Otherwise, just put [SVD.exe](#) into the same directory in which [lmgrd.exe](#) resides.

Now put the provided license file into a directory of this machine. Open the license file with a text editor and, if necessary, replace "*hostname*" by the correct hostname of this machine. To be specific, assume this hostname to be "[mycomputer](#)".

2.3 Start and Test the License Server

There are various possibilities to use the license server. For details please refer to file [LicenseAdministration.pdf](#). The following services-based possibility will usually do:

Configure the license server as a service:

- Start [lmtools.exe](#)².
- On the “Service/License File” tab, click the “Configuration using Services” option, and then click the “Config Services” tab.
Specify the name of the new service, e.g. “SVD License Manager”³, and path+name of [lmgrd.exe](#) and the SAMG license file, and specify path+name of a debug logfile⁴.
- Go to “Service/License File”, choose “Configuration using Services”, then choose the new service.
- Go to “Start/Stop/Reread” and start the new service.
- To test the SAMG-features: Go to “Server Diags”, specify “samg” as feature name and press the “Perform Diagnostics” button. FLEXlm should tell you “this license can be checked out”.
- Check the accessibility of the SAMG features additionally provided in your license file analogously.
- Note that, if you have replaced an old FLEXlm by a new one (see above), it might be necessary to specify, for instance, services also for your earlier FLEXlm-based licenses.



2.4 Providing the License

There is no need to copy the license file to all machines in your network.

Instead, on any machine in your network where a user wants to use SAMG, the user has to create, if not already existing, the environment variable [SVD_LICENSE_FILE](#) (rsp. [LM_LICENSE_FILE](#)), and let it point to (or include) the hostname of the license server, e.g. by issuing (for a console-based SAMG application prior to its call),



```
set SVD_LICENSE_FILE=@mycomputer
```

the license server’s hostname assumed to be “mycomputer”⁵. For permanent use, the user should add the above variable to Windows’ list of environment variables.

Remark: Since FLEXlm’s local memory of places where it found licenses, the Windows registry at: *HKEY_CURRENT_USER\SOFTWARE\FLEXlm License Manager*, might overstrike the definitions in the system environment variables, the user should delete the respective definitions in the registry-entry⁶ and set the system environment variable such that it contains the valid license, whenever the place containing the valid SAMG license has changed.



² On Windows Vista or later, you must run the lmtools utility as an administrator.

³ If you leave this field blank, the service will be named “FlexNet Publisher Service”.

⁴ Prepending the debug log file name with the + character appends logging entries. The default location for the debug log file is the “c:\ProgramData\FNP_DIR\debug.log” folder. To specify a different location, make sure you specify a fully qualified path. It is best practice to set the location of debug_log_path to a subfolder of “ProgramData”. This location has user-write permission by default, which is needed when the license server runs as a service with LocalService permission.

⁵ Under some conditions, there can be issues with resolving a host name. The user can also use @<ip-address – of-the-license-server-machine> directly, then.

⁶ Warning: In order not to affect other installed software, manually editing the registry requires great care!

3 Linux / MacOS

Below we describe steps to get the FLEXlm license server for SAMG started on Linux and MacOS. For more information on FLEXlm and its operation, we refer to the **FLEXlm's License Administration Guide** that is provided as part of SAMG's License Server archives for Linux resp. MacOS.

3.1 License Server Archive's Content

The [SVD_flexlm11.18.2.0](#) license server archives for Linux resp. MacOS contain the following files:

lmgrd	FLEXlm daemon, version 11.18.2.0
SVD	SAMG's vendor daemon; requires lmgrd v11.18.2.0 or higher
lmutil	Tool for starting, stopping, monitoring the license server etc.
LicenseAdministration.pdf	The original FLEXlm v11.18.2.0 License Administration Guide

3.2 Installation of the License Server

Locate the machine that is intended to run SAMG's license server (according to the data you specified to us). Depending on whether or not FLEXlm is already installed on that machine, do the following⁷:

- If FLEXlm is *not* yet installed, unpack the [SVD_flexlm11.18.2.0](#) archive for the respective OS to a directory of that machine.
- If FLEXlm *is* already installed, issue [lmutil lmver lmgrd](#) to find out its version number. Then do the following:
 - If the version is *older* than 11.18.2.0, replace [lmgrd](#) and [lmutil](#) by the corresponding newer versions provided in the archive and put [SVD](#) into the same directory in which [lmgrd](#) resides.
 - Otherwise, just put [SVD](#) into the same directory in which [lmgrd](#) resides.

Now put the license file provided – called "[yourlicensefile](#)" in the following – into a directory of this machine. Open the license file with a text editor and, if necessary, adapt its contents to your needs, e.g. replace "hostname" by the correct hostname of this machine. To be specific, assume this hostname to be "[machine1.mydomain.com](#)".

⁷ Here and in the following, we assume a bash shell.

3.3 Start and Test the License Server

There are various possibilities to use the license server. The following possibility will usually do:

- Go to the directory where `lmgrd` and `SVD` are located.
- To start the license server, issue

```
lmgrd -c yourlicensefile > flex.log 2>&1
```

This assumes that `yourlicensefile` is located in the same directory as `lmgrd` and `SVD`, and that the FLEXlm logfile shall be called `flex.log`.

- One of the last lines in the logfile `flex.log` should then resemble:

```
9:52:57 (SVD) Server started on machine1.mydomain.com for: samg
9:52:57 (SVD) <all features additionally provided in your license file>
```

If this straightforward approach should not be suitable to your needs, please refer to file [LicenseAdministration.pdf](#).

3.4 Providing the License

There is no need to copy the license file to all machines in your network. Instead, on any machine in your network where a user wants to use SAMG, the user has to create, if not already existing, the environment variable `SVD_LICENSE_FILE` (rsp. `LM_LICENSE_FILE`) and let it point to (or include) the hostname of the license server, e.g. by issuing, prior to calling an SAMG application:

```
export SVD_LICENSE_FILE=@machine1.mydomain.com
```

the license server's hostname assumed to be "`machine1.mydomain.com`"⁸. For permanent use, the user should add the above statement to `.bashrc` or a corresponding file.

Remark: Since FLEXlm's local memory of places where it found licenses, file `$HOME/.flexlmrc`, might overstrike the definitions in the system environment variables, the user should delete that local memory whenever a new license is to be used and set the system environment variable such that it contains the valid license.



⁸ Under some conditions, there can be issues with resolving a host name. The user can also use `@<ip-address – of-the-license-server-machine>` directly, then.